

Strategies for Reducing Body Water (Drying Out)



STRATEGIES FOR REDUCING BODY WATER (DRYING OUT)*

The following educational content may improve appearance only if excess water retention is present but is not recommended. All athletes should consult the appropriate medical professional before attempting the techniques described here. The information presented in this table is exclusively for educational purposes.

Strategy	Mechanism	Example Strategy
Water load	Reduce blood $[Na^+]$ diuresis	With a constant rate or total intake of Na^+ , an athlete consumes 25 to 50% above habitual rate/ total water intake, giving 8 to 12 hours to permit diuresis. Simply reducing Na^+ intake and maintaining the rate of water intake will create the same effect.
Na^+ intake reduction	Reduce blood $[Na^+]$	Maintain water intake, but reduce habitual sodium intake to that of low sodium foods.
Increased protein intake	Increase protein intake to generate urea and promote diuresis in the kidney	Maintain or increase protein intake ($\sim 3g / kg$ of LBM/day) the day before the competition to promote diuresis.
Reduced carbohydrate intake	Reduced carbohydrate with a ketogenic diet to reduce body water through various potential mechanisms	Minimize carbohydrates, increase, protein intake, and consume adequate fats consumed to meet caloric needs or restore muscle triglycerides that may have been reduced during the preceding carbohydrate loading period.
Caffeine	Increase renal filtration, reduce sodium conservation, and inhibit hepato-renal reflex	~ 250 mg dose (as a caffeinated beverage or otherwise) or the equivalent above habitual intake the day before the competition, but early enough to avoid sleep disturbance.
Vitamin C	Excrete osmolyte	Consume dose of 500 to 1,000 mg every 3 hours after diuresis are initiated Be cautious of gastric distress/osmotic diarrhea.

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Head-down tilt	Release a natriuretic factor from the heart causing diuresis	After initiating diuresis, the athlete rests and sleeps on a downward-sloping surface, typically a bed/couch, the night before the competition. A 6+° downward angle (e.g., raising the foot of the bed by 6 to 12 inches) should not cause discomfort, such as blood rushing to the head, sliding down the bed, or disorientation. This approach is not suggested in those with a tendency toward gastric reflux. Head-down tilt would be employed through the competitive days as needed to encourage diuresis and a dry condition.
Stress management	Mediate water retention via various mechanisms	Anecdotal and scientific evidence suggests that psychological stress can impact water retention. Strategies such as performing one or more peak-week trial run to increase confidence and minimize guesswork before competing, staying busy and entertained during peak week, listening to music, meditation, and hiring a coach may all be applicable. An individual solution to stress management is best developed as early as possible, including during trial runs, so that the athlete can begin developing skills long before the competition/photoshoot to address stress preventatively and proactively.